

Cycle stock: batch size, cycle frequency and average inventory

Part 2 · Categories of stock — solution report with data, method, results and charts

Contents

1. Part 2 · Categories of stock — Cycle stock: batch size, cycle frequency and average inventory

1. Cycle stock: batch size, cycle frequency and average inventory

Annual demand $D = 12,000$ units over 240 working days $\rightarrow$ demand rate $d = 50.0$ units/day.

$$\text{Average cycle stock} = Q / 2 = D / (2n)$$

4,000 units $\rightarrow$ avg 2,000

Largest lot (Q)

50.0 units/day

Demand rate

3 $\rightarrow$ 6 cycles

Doubling cycles/year

50 %

Avg cycle stock falls by

Saw-tooth cycle stock: smaller, more frequent lots cut average inventory

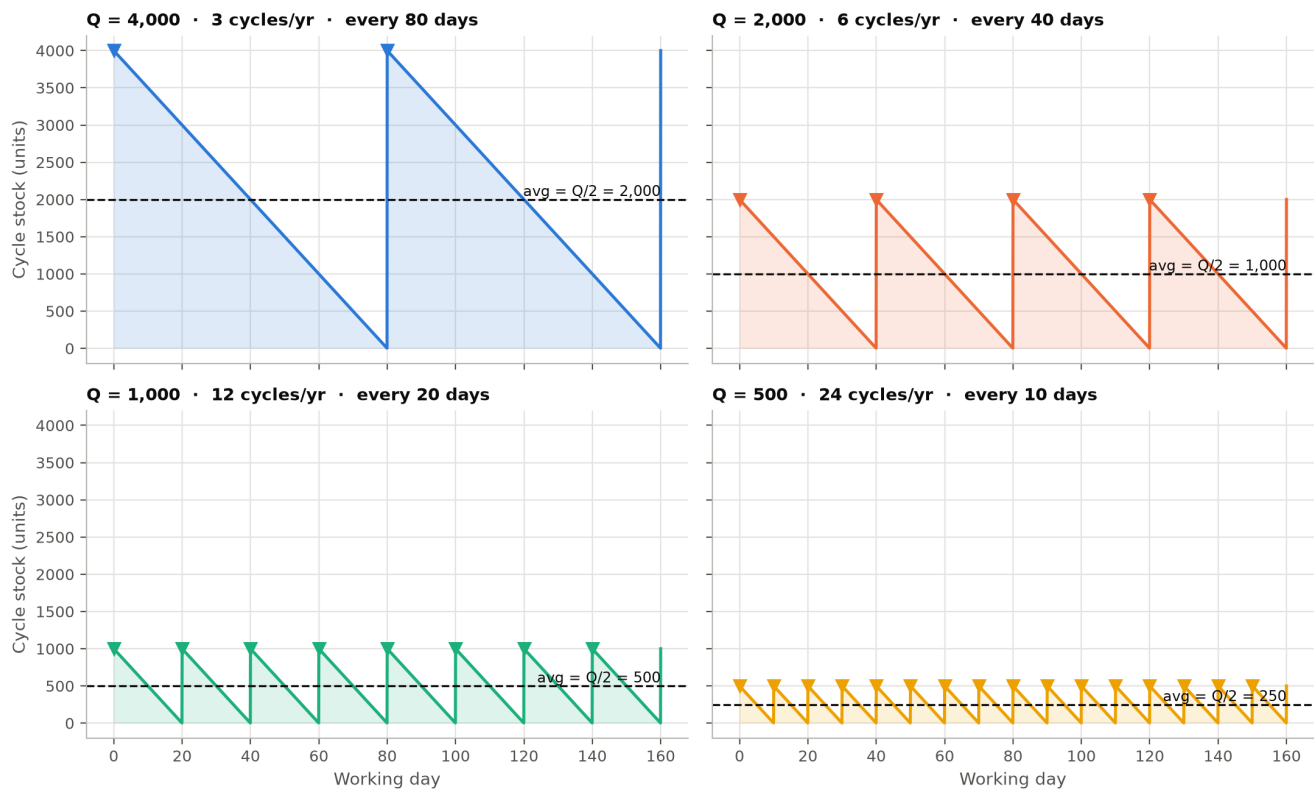


Figure 1.1 — Cycle-stock profiles for the four lot sizes. ▼ marks a make/order event; dashed line = average ($Q/2$).

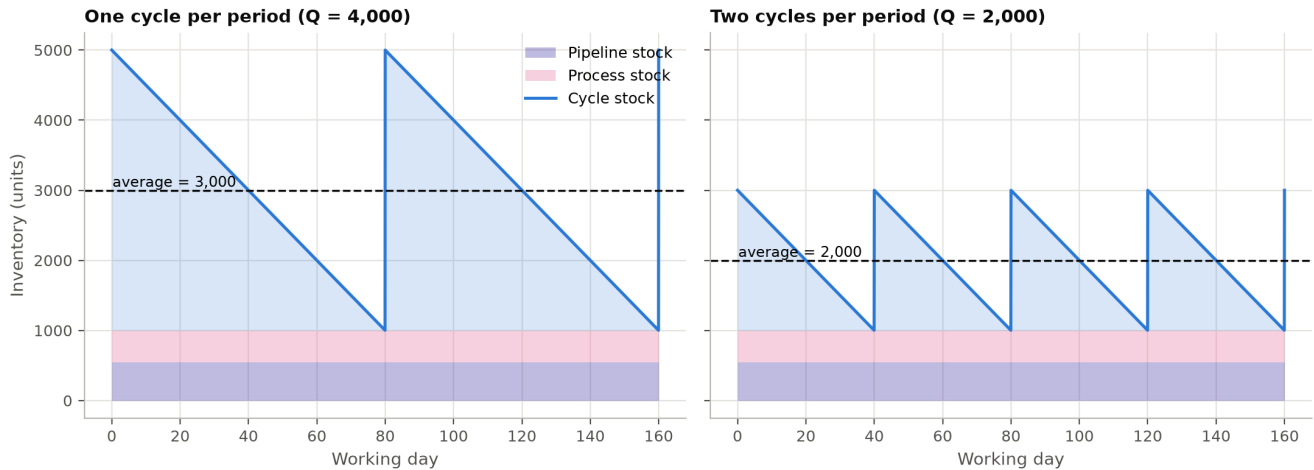
Doubling the number of cycles halves average cycle stock (fixed base stock unchanged)

Figure 1.2 — Replica of lecture slide 19 with pipeline and process stock as a fixed base.

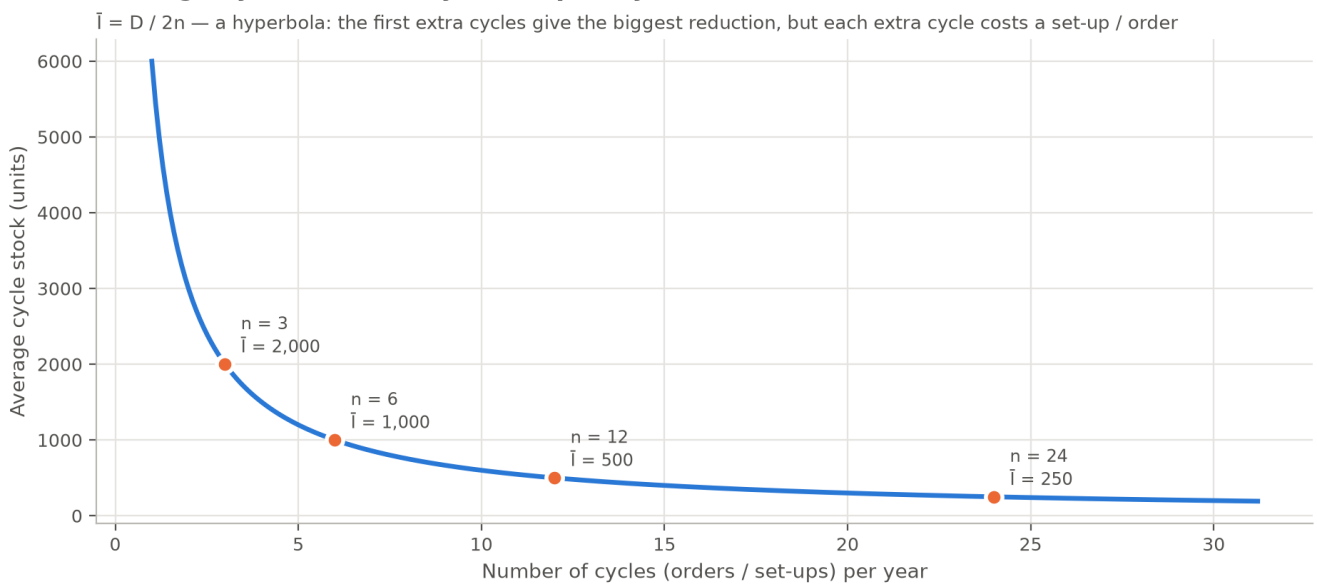
Average cycle stock vs cycle frequency

Figure 1.3 — Trade-off preview: fewer units in stock requires more frequent set-ups (see Part 6.4).

Cycle stock for alternative lot sizes.

lot size Q	cycles per year	cycle time days	max cycle stock	min cycle stock	avg cycle stock units	avg coverage days
4,000	3.0	80.0	4,000	0	2,000	40.0
2,000	6.0	40.0	2,000	0	1,000	20.0
1,000	12.0	20.0	1,000	0	500	10.0
500	24.0	10.0	500	0	250	5.0